

Java JSP (JavaServer Pages) - Full Notes

What is JSP?

JSP (JavaServer Pages) is a server-side technology used to create dynamic web pages. It allows mixing HTML and Java code in one file.

Features of JSP

- Mix of HTML and Java code
- Easier and more readable than Servlets
- Reusable and maintainable
- Internally converted to a Servlet

Life Cycle of JSP

1. Translation: JSP is converted to a Servlet.
2. Compilation: The Servlet is compiled.
3. Instantiation: Object of JSP is created.
4. Initialization (jspInit()): Called once.
5. Request Handling (_jspService()): Called for every request.
6. Destroy (jspDestroy()): Called at end for cleanup.

Types of JSP Tags

<% ... %> : Scriptlet - for Java code

<%= ... %> : Expression - for output

<%! ... %> : Declaration - for variables/methods

<%@ ... %> : Directive - for instructions

Example JSP Page

```
<%@ page language="java" contentType="text/html" %>
```

```
<html>
```

```
<head><title>JSP Example</title></head>
```

```
<body>
```

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```
<%  
    String name = "Deepa";  
%>  
  
<h1>Hello <%= name %>!</h1>  
  
</body>  
  
</html>
```

Directive Tags

1. page directive: Gives info about the page

```
<%@ page language="java" contentType="text/html" %>
```

2. include directive: Includes another file

```
<%@ include file="header.jsp" %>
```

3. taglib directive: Includes custom tag libraries

```
<%@ taglib uri="..." prefix="..." %>
```

JSP vs Servlet

Servlet:

- Java code in HTML
- Hard to read/maintain
- Logic-oriented

JSP:

- HTML with Java code
- Easier to read
- Presentation-oriented

Implicit Objects in JSP

request: Handles client request

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response: Sends response to client

out: Writes output to client

session: Holds user session data

application: Stores data in application scope

config: Configuration of JSP

pageContext: Controls entire JSP page

Why Use JSP?

- Easy for designers
- Separates presentation and logic (MVC)
- Reduces server load
- Useful for dynamic websites

Handling Form Data in JSP

```
<form method="post" action="submit.jsp">  
    Name: <input type="text" name="username">  
    <input type="submit" value="Submit">  
</form>
```

submit.jsp:

```
<%  
    String user = request.getParameter("username");  
%>  
<h2>Hello <%= user %>!</h2>
```